Comparison Between the Secrecy Capacity of Reconfigurable Intelligent Surface Aided and Relay-Assisted Wireless Communication Systems in the Presence of Multiple Eavesdroppers

***Abstract*—This study presents a comparative analysis of the secrecy performance for RIS-assisted and relay-assisted wireless communication systems in a worst-case multiple-eavesdropper scenario. We statistically characterize the end-to-end communication link to obtain novel closed-form expressions for the ergodic secrecy capacity (ESC) of both systems and investigate the effects of key parameters such as number of RIS-reflecting elements, number of eavesdroppers, and signal-to-noise ratio (SNR). The analytical results are checked through extensive Monte Carlo simulations which are found to be in close agreement with the analytical results. The results indicate that the number of RIS-reflecting elements significantly enhances the ESC while the number of eavesdroppers in both systems deteriorates the secrecy performance. Furthermore, the RIS-aided system shows better performance than the relay-aided system in all network conditions considered, thereby highlighting the capability of RIS as a powerful and energy-efficient system for physical-layer security in future wireless networks.**

***Index Terms*—Reconfigurable intelligent surface (RIS), relay-assisted communications, physical layer security (PLS), ergodic secrecy capacity (ESC), multiple eavesdroppers, wireless communications.**

I. Introduction

Wireless communication has become an important part of modern networks. However, its open nature makes confidential information vulnerable to eavesdropping. Although traditional encryption techniques provide security, they rely on complex computations and secure key management. To overcome these challenges, physical layer security (PLS) has become an effective approach for securing 5G and 6G networks. Reconfigurable intelligent surface (RIS) is one of the available technologies that has attracted significant research interest because of its capability of controlling the propagation of wireless signals, enhancing the quality of legitimate links, and reducing information leakage with low power consumption. In comparison, relay-assisted communication improves coverage and reliability but requires extra hardware and transmit power. Therefore, RIS-assisted communication provides a simple, low-cost and energy-efficient solution for secure wireless communication.

A. Related Works

The application of reconfigurable intelligent surfaces (RISs) has attracted significant research interest due to their capability to enhance the reliability, capacity, and security of wireless communication systems. Boulogeorgos and Alexiou [1] developed a comprehensive analytical framework comparing RIS-aided and relay-assisted wireless systems by deriving the expression for key performance metrics. Their results demonstrated the superiority of RIS-assisted transmission; however, physical layer security was not considered. Relay-assisted physical layer security has also been extensively studied in the literature. Zhao et al. [2] derived analytical expressions for the SOP and ESC of relay systems over generalized K fading channels and further presented asymptotic analyses together with relay selection strategies. Kalamkar and Banerjee [3] analyzed the secrecy performance of energy-harvesting untrusted relay systems. However, their work neither considered RIS-assisted transmission nor provided a comparative secrecy analysis between RIS and relay-aided systems. With the emergence of RIS technology, several studies have explored its application to secure wireless communications. Wei et al. [4] examined the secure communication capability of RIS-assisted systems by considering randomly distributed UAV eavesdroppers and derived closed-form equations for the ergodic secrecy capacity. Khalid et al. [5] investigated RIS-assisted wireless powered communication and obtained mathmatical equations for outage probability and secrecy capacity, demonstrating the impact of RIS on both communication reliability and security. Yi et al. [6] proposed a RIS-aided seamless connectivity framework for wireless multi-hop relay networks with RIS-equipped UAV communication nodes and demonstrated that seamless RIS-assisted transmission significantly improves network performance through enhanced diversity gain. More recently, RIS-assisted cooperative communication architectures have also received considerable attention. Chen et al. [7] proposed a multi-RIS-assisted cooperative NOMA framework and analyzed its reliability and security performance by evaluating outage probability, ergodic capacity, intercept probability, detection error probability, and system security probability. Similarly, Mensi and Rawat [8] evaluated partial RIS selection against partial relay selection in vehicular communication systems and demonstrated that RIS-assisted transmission achieves improved secrecy capacity and communication performance compared with conventional relay-assisted schemes. Shahriyer et al. [9] evaluated the PLS performance of cooperative relay-

based wireless multicast networks employing opportunistic relay selection over generalized κ–μ shadowed channels. They derived several performance metrics, including PNZSC, SOP, and ESC, along with asymptotic analyses to evaluate the impact of fading parameters and multiple eavesdroppers. However, their work focused on relay-assisted multicast networks and did not consider RIS-assisted transmission or a comparative secrecy analysis between RIS-assisted and relay-assisted wireless systems. Despite these contributions, a comprehensive secrecy comparison between RIS-assisted and relay-assisted wireless systems under a worst-case multiple-eavesdropper scenario remains unexplored. Existing studies mainly focus on communication reliability and performance metrics, with limited attention to ergodic secrecy capacity. To address this gap, this paper derives closed-form ESC expressions for RIS-assisted and relay-assisted wireless systems under multiple eavesdroppers and validates the analysis through Monte Carlo simulations.

B. Contributions

Motivated by the above observations, this work investigates the comparative secrecy capability of RIS-aided and relay-aided wireless communication systems considering multiple eavesdroppers. The proposed framework offers the following key features:

- A mathematical framework is formulated to investigate the ergodic secrecy capacity of RIS-aided and relay-aided transmission strategies under multi-eavesdropper conditions.
- Exact analytical formulations of the ergodic secrecy capacity are established for both transmission schemes over Rayleigh fading channels, and the theoretical results are validated through numerical simulations.
- The proposed analysis provides quantitative insights into how the RIS size, the RIS-reflecting elements, the number of eavesdroppers, and the average SNR of the legitimate link influence the achievable ergodic secrecy capacity.

The remainder of this paper is organized as follows. Section II describes the system model and channel assumptions. The analytical derivation of the secrecy metrics is presented in Section III. Section IV provides numerical evaluations along with simulation validation and Section V concludes the study.

II. System Model

The considered network comprises a base station (BTS), a legitimate destination (D), and multiple eavesdroppers (E), where the BTS transmits confidential information to the destination. The direct LOS paths from the BTS to the legitimate receiver as well as the eavesdroppers are assumed to be blocked. Under these assumptions, two transmission scenarios are investigated.

A. BTS–RIS–Destination and BTS–RIS–Eavesdropper Links

Consider a wireless communication scenario in which the direct transmission paths towards the destination and multiple

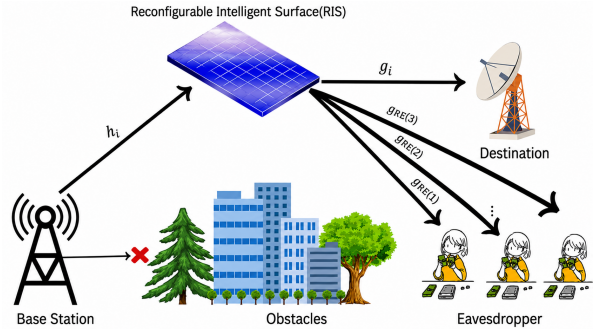


Fig. 1. System model of the proposed scheme.

eavesdroppers are blocked. Consequently, communication is assisted by an RIS comprising N passive reflecting elements, as shown in Fig. 1. The RIS reflects the transmitted signal toward the legitimate destination, while the reflected signal may also be intercepted by multiple eavesdroppers. The signal received at the i-th RIS reflecting element is given by

$$r_i = h_i s + n_{R,i}, \qquad i = 1, 2, \ldots, N, \tag{1}$$

where s denotes the transmitted signal satisfying $\mathbb{E}[|s|^2] = P_s$, h_i represents the fading coefficient associated with the communication link connecting the BTS with the i-th RIS reflecting element and $n_{R,i} \sim \mathcal{CN}(0, N_0)$ is the noise component at the RIS.

Each RIS element applies an adjustable phase shift, denoted by θ_i, before reflecting the incident signal. Accordingly, the signal received at the destination can be expressed as

$$y_D = \sum_{i=1}^{N} g_i e^{j\theta_i} \left(h_i s + n_{R,i}\right) + n_D = \Gamma_D s + \nu_D, \tag{2}$$

where g_i represents the fading coefficient associated with the wireless path linking the i-th RIS reflecting element to the destination, while $n_D \sim \mathcal{CN}(0, N_0)$ represents the noise component at the destination, and

$$\Gamma_D = \sum_{i=1}^{N} g_i e^{j\theta_i} h_i \tag{3}$$

corresponds to the effective cascaded channel gain of the legitimate communication link.

Assume that Q passive eavesdroppers attempt to intercept the reflected transmission. The received signal at the q-th eavesdropper is given by

$$y_{E_q} = \sum_{i=1}^{N} g_{RE(q),i} e^{j\theta_i} \left(h_i s + n_{R,i}\right) + n_{E_q}. \tag{4}$$

where $q = 1, 2, \ldots, Q$ denotes the index of the eavesdroppers, $g_{RE(q),i}$ indicates the channel coefficient between the i-th RIS element and the q-th eavesdropper, and $n_{E_q} \sim \mathcal{CN}(0, N_0)$ represents the corresponding AWGN. The equivalent cascaded channel for the q-th eavesdropper is

$$\Gamma_{E_q} = \sum_{i=1}^{N} g_{RE(q),i} e^{j\theta_i} h_i. \tag{5}$$

B. BTS–Relay–Destination and BTS–Relay–Eavesdropper Links

For the second transmission scenario, communication is supported by the relay that forwards the confidential signal from the base station (BTS) to the legitimate destination, as illustrated in Fig. 1. As the direct communication paths from the BTS to both the legitimate destination and the eavesdroppers are assumed to be unavailable, all information is forwarded through the relay. The relay operates in half-duplex mode, indicating that it cannot transmit and receive simultaneously. Instead, it first receives the transmitted signal, amplifies it with a fixed gain, and retransmits the amplified signal to the destination, while multiple eavesdroppers may also intercept the transmission.

The signal received at the relay is expressed as

$$z_R = f\, s + n_R. \tag{6}$$

where f represents the channel coefficient between the BTS and the relay, and n_R represents the AWGN introduced at the relay.

After amplification with a fixed relay gain G, the forwarded signals at the destination and the q-th eavesdropper can be expressed as

$$y_D = c_D G(f\, s + n_R) + n_D = \Psi_D s + \omega_D, \tag{7}$$

$$y_{E_q} = c_{E_q} G(f\, s + n_R) + n_{E_q} = \Psi_{E_q} s + \omega_{E_q}. \tag{8}$$

where $q = 1, 2, \ldots, Q$. The propagation coefficients corresponding to the relay-destination link and the relay–q-th eavesdropper link are denoted by c_D and c_{E_q}, respectively. In addition, n_D and n_{E_q} denote the receiver noise at the destination and the q-th eavesdropper, respectively. The effective end-to-end channel coefficients are defined as

$$\Psi_D = c_D G f, \tag{9}$$

and

$$\Psi_{E_q} = c_{E_q} G f, \qquad q = 1, 2, \ldots, Q. \tag{10}$$

III. Problem Formulation

This section presents the secrecy performance analysis of two wireless communication architectures: (i) an RIS-assisted communication system and (ii) a conventional relay-assisted communication system. In both scenarios, the base station (BTS) aims to securely transmit confidential information to the legitimate destination (D) while multiple eavesdroppers (E) try to decode the confidential information. It is assumed that all wireless links undergo Rayleigh fading and no direct line-of-sight (LOS) path exists between the BTS and either the destination or the eavesdropper(s).

A. RIS-aided wireless communication system

The probability density function (PDF) of the received SNR at the legitimate receiver is given by

$$f_{\gamma_d}(\gamma_d) = \frac{1}{2b_m\, P_s^{\frac{a_m+1}{2}} \Gamma(a_m+1)} \gamma_d^{\frac{a_m-1}{2}} \exp\left(-\frac{1}{b_m}\sqrt{\frac{\gamma_d}{P_s}}\right). \tag{11}$$

for $\gamma_d > 0$.

where the constants are defined as

$$A_1 = \frac{1}{2b_m P_s^{\frac{a_m+1}{2}} \Gamma(a_m+1)},\ A_2 = \frac{a_m-1}{2},\ A_3 = \frac{1}{b_m\sqrt{P_s}}.$$

The PDF of the received SNR at the eavesdropper is given by

$$f_{\gamma_{e_j}}(\gamma_{e_j}) = \frac{1}{2b_e\, P_s^{\frac{a_e+1}{2}} \Gamma(a_e+1)} \gamma_{e_j}^{\frac{a_e-1}{2}} \exp\left(-\frac{1}{b_e}\sqrt{\frac{\gamma_{e_j}}{P_s}}\right). \tag{12}$$

for $\gamma_{e_j} > 0$.

where the constants are defined as

$$B_1 = \frac{1}{2b_e P_s^{\frac{a_e+1}{2}} \Gamma(a_e+1)},\ B_2 = \frac{a_e-1}{2},\ B_3 = \frac{1}{b_e\sqrt{P_s}}.$$

The Ergodic Secrecy Capacity is expressed as

$$\langle C_s^{\mathrm{RIS}} \rangle = \int_0^\infty \log_e(1+\gamma_d) f_{\gamma_d}(\gamma_d)\, d\gamma_d - \int_0^\infty \log_e(1+\gamma_e) f_{\gamma_{\max}}(\gamma_e)\, d\gamma_e \tag{13}$$

Using the identities from [10] (3.351.1) and (8.352.6), the closed-form equation of ESC can be written as

$$\begin{aligned}\langle C_s^{\mathrm{RIS}} \rangle = {} & \frac{2A_1}{A_3^{2A_2+2}} \sum_{n_1=1}^{\infty} \frac{(-1)^{n_1+1}}{n_1} \left(\frac{1}{A_3^2}\right)^{n_1} (2n_1 + 2A_1 + 1)! \\ & - Q\, \frac{2B_1}{B_3^{2B_2+2}} \sum_{n_2=1}^{\infty} \frac{(-1)^{n_2+1}}{n_2} \left(\frac{1}{B_3^2}\right)^{n_2} \\ & \times (2B_2 + 2n_2 + 1)! \left[\frac{2B_1}{B_3^{2B_2+2}} (2B_2+1)!\right]^{Q-1} \\ & \times \left[1 - \exp\left(-\left(\frac{x}{B_3}\right)^2\right) \sum_{m=0}^{2B_2+1} \frac{x^{2m}}{B_3^{2m} m!}\right]^{Q-1}.\end{aligned} \tag{14}$$

B. Relay-assisted wireless communication system

The PDF of received SNR at the destination user is expressed as

$$\begin{aligned} f_{\gamma_m}(\gamma_m) = {} & \frac{2}{\bar{P}_s \bar{P}_n}\, \gamma_m \exp\left[-\left(\frac{1}{\bar{P}_s} + \frac{1}{\bar{P}_n}\right)\gamma_m\right] \\ & \times \left\{ \frac{\bar{P}_s + \bar{P}_n}{\sqrt{\bar{P}_s \bar{P}_n}} K_1\left(\frac{2\gamma_m}{\sqrt{\bar{P}_s \bar{P}_n}}\right) + 2K_0\left(\frac{2\gamma_m}{\sqrt{\bar{P}_s \bar{P}_n}}\right)\right\}. \end{aligned} \tag{15}$$

$$f_{\gamma_m}(\gamma_m) = A_1 A_3 \sqrt{\frac{\pi}{2}}\, e^{-\gamma_m (A_2+A_4)} \sum_{r_1=0}^{\infty} \frac{2^{-r_1}\left(r_1-\frac{1}{2}\right)!}{r_1!\left(\frac{1}{2}-r_1\right)!} \times \gamma_m^{-1-r_1} A_4^{-\left(\frac{1}{2}+r_1\right)} + A_1\sqrt{2\pi}\, e^{-\gamma_m (A_2+A_4)} \sum_{r_0=0}^{\infty} \frac{2^{-r_0}\left(r_0-\frac{1}{2}\right)!}{r_0!\left(-r_0-\frac{1}{2}\right)!} \times \gamma_m^{\frac{1}{2}-r_0} A_4^{-\left(\frac{1}{2}+r_0\right)}. \tag{16}$$

where the constants are defined as,

$$A_1 = \frac{2}{\bar{P}_s \bar{P}_n}, \qquad A_2 = \frac{1}{\bar{P}_s} + \frac{1}{\bar{P}_n},$$

$$A_3 = \frac{\bar{P}_s + \bar{P}_n}{\sqrt{\bar{P}_s \bar{P}_n}}, \qquad A_4 = \frac{2}{\sqrt{\bar{P}_s \bar{P}_n}}.$$

The PDF of the received SNR at the Eavesdropper can be written as

$$f_{\gamma_{E_j}}(\gamma_{E_j}) = \frac{2}{\bar{P}_s \bar{P}_r}\, \gamma_{E_j} \exp\left[-\left(\frac{1}{\bar{P}_s}+\frac{1}{\bar{P}_r}\right)\gamma_{E_j}\right] \times \left\{ \frac{\bar{P}_s+\bar{P}_r}{\sqrt{\bar{P}_s \bar{P}_r}} K_1\left(\frac{2\gamma_{E_j}}{\sqrt{\bar{P}_s \bar{P}_r}}\right) + 2K_0\left(\frac{2\gamma_{E_j}}{\sqrt{\bar{P}_s \bar{P}_r}}\right)\right\}. \tag{17}$$

$$f_{\gamma_{E_j}}(\gamma_{E_j}) = A_1 A_3 \sqrt{\frac{\pi}{2}}\, e^{-\gamma_{E_j}(A_2+A_4)} \sum_{r_1=0}^{\infty} \frac{2^{-r_1}\left(r_1+\frac{1}{2}\right)!}{r_1!\left(\frac{1}{2}-r_1\right)!} \times \gamma_{E_j}^{\frac{1}{2}-r_1} A_4^{-\left(\frac{1}{2}+r_1\right)} + A_1\sqrt{2\pi}\, e^{-\gamma_{E_j}(A_2+A_4)} \sum_{r_0=0}^{\infty} \frac{2^{-r_0}\left(r_0-\frac{1}{2}\right)!}{r_0!\left(-r_0-\frac{1}{2}\right)!} \times \gamma_{E_j}^{\frac{1}{2}-r_0} A_4^{-\left(\frac{1}{2}+r_0\right)}. \tag{18}$$

The Ergodic Secrecy Capacity is expressed as

$$\left\langle C_s^{\text{Relay}} \right\rangle = \int_0^{\infty} \log_e(1+\gamma_m) f_{\gamma_m}(\gamma_m)\, d\gamma_m - \int_0^{\infty} \log_e(1+\gamma_{E_j}) f_{\gamma_{E_{\max}}}(\gamma_{E_j})\, d\gamma_{E_j}. \tag{19}$$

Using the identities from [10] (3.351.1), (8.352.6), and (3.351.3), the closed-form equation of ESC can be written as

$$\begin{aligned}\left\langle C_s^{\text{Relay}} \right\rangle = & \sum_{n_1=1}^{\infty} \frac{(-1)^{n_1+1}}{n_1} \left(\frac{1}{2}-r_1+n_1\right)! \times (A_2+A_4)^{-\frac{3}{2}+r_1-n_1} A_1\sqrt{\pi} \sum_{r_1=0}^{\infty} \frac{2^{-r_1}}{r_1!} \times A_4^{-\left(\frac{1}{2}+r_1\right)} \left[\frac{A_3}{\sqrt{2}} \sum_{r_1=0}^{\infty} \frac{\left(r_1+\frac{1}{2}\right)!}{\left(\frac{1}{2}-r_1\right)!} + \sqrt{2} \sum_{r_1=0}^{\infty} \frac{\left(r_1-\frac{1}{2}\right)!}{\left(-r_1-\frac{1}{2}\right)!}\right] \\ & - C\left(\frac{1}{2}-r_1+n_1\right)! (A_2+A_4)^{-\frac{3}{2}+r_1-n_1} - D(1-r_0-r_1+n_2)! \left[2(A_2+A_4)\right]^{-n_2-1} + E(1-r_0-r_1+n_2+m)! \left[3(A_2+A_4)\right]^{-n_2-1} \\ & - F\left(\frac{1}{2}-r_0+n_3\right)! (A_2+A_4)^{-n_3-1} + G\left(\frac{1}{2}-r_0+m+n_3\right)! \left[2(A_2+A_4)\right]^{-n_3-1}.\end{aligned} \tag{20}$$

where,

$$D = A_1 A_3 \sqrt{\frac{\pi}{2}} + A_1\sqrt{2\pi} \sum_{n_2=1}^{\infty} \frac{(-1)^{n_2+1}}{n_2} \times \sum_{r_0=0}^{\infty} \frac{2^{-r_0}\left(r_0-\frac{1}{2}\right)!}{r_0!\left(-r_0-\frac{1}{2}\right)!} A_4^{-\left(\frac{1}{2}+r_0\right)} \times \sum_{r_1=0}^{\infty} A_4^{-\left(\frac{1}{2}+r_1\right)} \frac{2^{-r_1}\left(r_1+\frac{1}{2}\right)!}{r_1!}. \tag{21}$$

$$E = A_1 A_3 \sqrt{\frac{\pi}{2}} + A_1\sqrt{2\pi} \sum_{n_2=1}^{\infty} \frac{(-1)^{n_2+1}}{n_2} \times \sum_{r_0=0}^{\infty} \frac{2^{-r_0}\left(r_0-\frac{1}{2}\right)!}{r_0!\left(-r_0-\frac{1}{2}\right)!} A_4^{-\left(\frac{1}{2}+r_0\right)} \times \sum_{r_1=0}^{\infty} \frac{2^{-r_1}\left(r_1+\frac{1}{2}\right)!}{r_1!} A_4^{-\left(\frac{1}{2}+r_1\right)} \sum_{m=0}^{\frac{1}{2}-r_1} \frac{\left(\frac{1}{2}-r_1\right)^m}{m!} \tag{22}$$

$$F = A_1^2\, 2\pi \sum_{r_0=0}^{\infty} \frac{2^{-r_0}\left(r_0-\frac{1}{2}\right)!}{r_0!\left(-r_0-\frac{1}{2}\right)!} A_4^{-2\left(r_0+\frac{1}{2}\right)} \times (A_2+A_4)^{-\frac{3}{2}+r_0} \left(\frac{1}{2}-r_0\right)! \sum_{n_3=1}^{\infty} \frac{(-1)^{n_3+1}}{n_3}. \tag{23}$$

$$G = A_1^2\, 2\pi \sum_{r_0=0}^{\infty} \frac{2^{-r_0}\left(r_0-\frac{1}{2}\right)!}{r_0!\left(-r_0-\frac{1}{2}\right)!} A_4^{-2\left(r_0+\frac{1}{2}\right)} \times (A_2+A_4)^{-\frac{3}{2}+r_0} \left(\frac{1}{2}-r_0\right)! \sum_{n_3=1}^{\infty} \frac{(-1)^{n_3+1}}{n_3} \times \sum_{m=0}^{\infty} \frac{(A_2+A_4)^m}{m!}. \tag{24}$$

IV. Numerical Results

The analytical ESC expression derived in Section III is evaluated numerically using Mathematica and subsequently plotted in MATLAB and verified via Monte Carlo simulations.

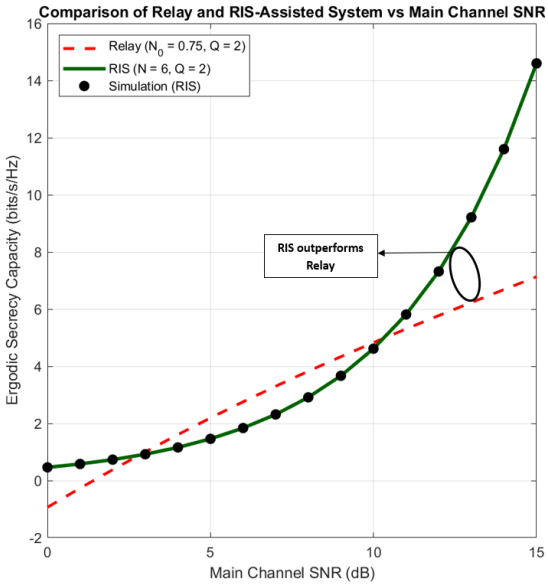


Fig. 2. Comparison of relay- and RIS-assisted systems in terms of ESC for $N = 6$ and $Q = 2$

Figure 2 shows the ergodic secrecy capacity (ESC) versus the main channel SNR for RIS-assisted and relay-assisted systems. Although the relay-assisted system performs slightly better at low SNR, the RIS-assisted system achieves higher ESC as the SNR increases. Simulation results closely match the theoretical analysis, thereby verifying the correctness of the proposed closed-form expressions.

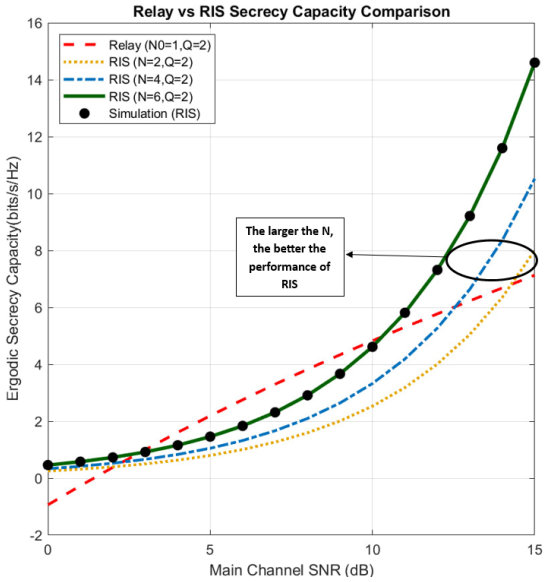


Fig. 3. Impact of the number of RIS reflecting elements (N) on ESC for $Q = 2$

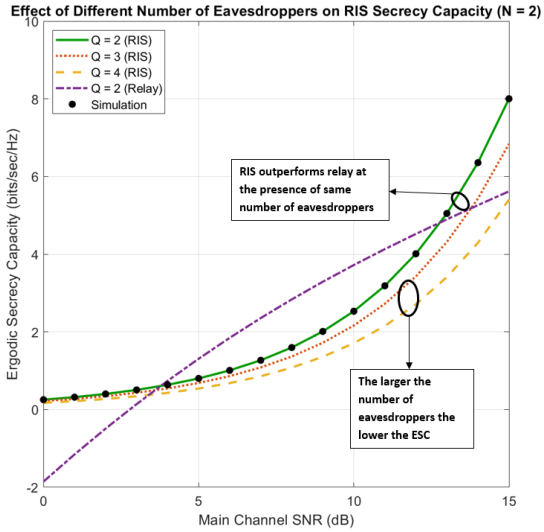


Fig. 4. Effect of relay noise power (N_0) on ESC for $Q = 2$.

Figures 3 and 4 present the ergodic secrecy capacity (ESC) versus the main channel SNR for different RIS configurations and numbers of eavesdroppers. As the main channel SNR increases, the ESC improves for all considered cases. Increasing RIS-reflecting elements from $N = 2$ to $N = 6$ substantially enhances the secrecy performance, with $N = 6$ achieving the highest ESC. In contrast, increasing the number of eavesdroppers from $Q = 2$ to $Q = 4$ degrades the ESC due to the increased interception capability. Moreover, for the same number of eavesdroppers, the RIS-enabled system consistently outperforms the relay-aided scheme.

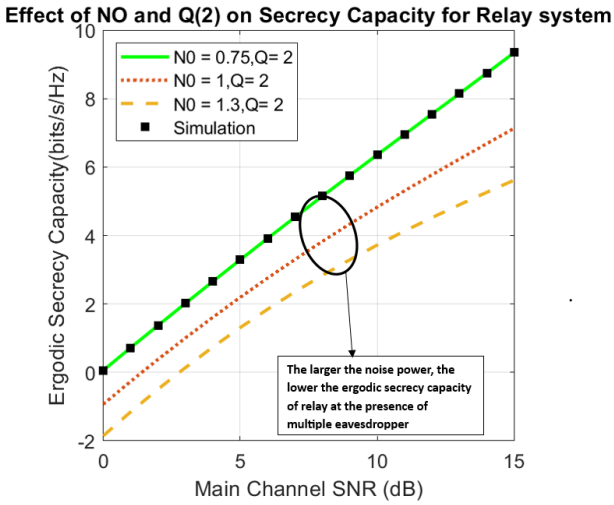


Fig. 5. Impact of the number of RIS reflecting elements (N) and eavesdroppers (Q) on ESC

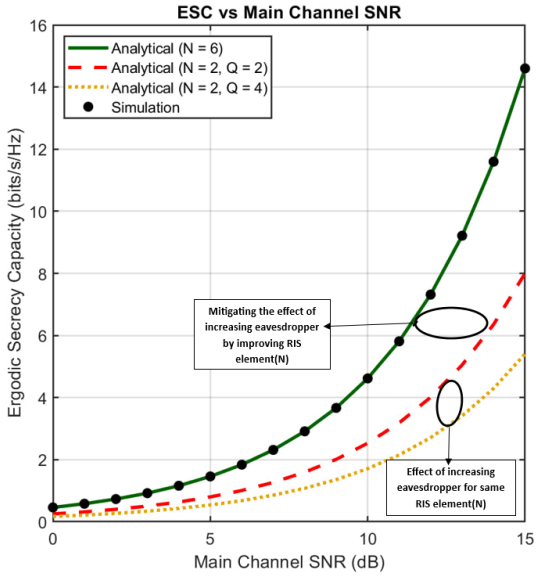


Fig. 6. Impact of the number of eavesdroppers (Q) on ESC for $N = 2$.

Figures 5 and 6 illustrate the impact of noise power and different eavesdropper configurations on the ergodic secrecy capacity (ESC). As the main channel SNR increases, the ESC improves for all considered cases. However, increasing the noise power from $N_0 = 0.75$ to $N_0 = 1.3$ degrades the secrecy performance with $N_0 = 0.75$ achieving the highest ESC. Similarly, increasing the number of eavesdroppers from $Q = 2$ to $Q = 3$ reduces the ESC due to stronger interception capability. This reduction in performance can be overcome by deploying additional RIS-reflecting elements where $N = 6$ provides a noticeable improvement over smaller RIS configurations. The Monte Carlo simulation results closely match the analytical curves, confirming the accuracy of the derived ESC expression.

V. Conclusion

In this study, the ergodic secrecy capacity (ESC) of RIS-assisted wireless communication systems and AF relay-assisted systems with multiple eavesdroppers has been compared. A set of closed-form expressions for the ESC was derived and validated through simulation, showing excellent agreement between the analytical and simulated results. The results show that the RIS-aided system can provide much higher secrecy capacity at higher SNR while the AF relaying system has slightly higher secrecy capacity at lower SNR. Furthermore, the more reflecting elements RIS has, the more secrecy performance it can achieve and the less impact it will have on multiple eavesdroppers. The results of this study underscore that RIS-aided communication can greatly contribute to physical layer security in the future wireless networks.

References

[1] Alexandros-Apostolos A Boulogeorgos and Angeliki Alexiou. Performance analysis of reconfigurable intelligent surface-assisted wireless systems and comparison with relaying. *IEEE Access*, 8:94463–94483, 2020.

[2] Hui Zhao, Zhedong Liu, Liang Yang, and Mohamed-Slim Alouini. Secrecy analysis in df relay over generalized-k fading channels. *IEEE Transactions on Communications*, 67(10):7168–7182, 2019.

[3] Sanket S Kalamkar and Adrish Banerjee. Secure communication via a wireless energy harvesting untrusted relay. *arXiv preprint arXiv:1509.08262*, 2015.

[4] Lai Wei, Kezhi Wang, Cunhua Pan, and Maged Elkashlan. Secrecy performance analysis of ris-aided communication system with randomly flying eavesdroppers. *IEEE Wireless Communications Letters*, 11(10):2240–2244, 2022.

[5] Waqas Khalid, Manish Nair, Trinh Van Chien, and Heejung Yu. Impact of ris on outage probability and ergodic rate in wireless powered communication. *arXiv preprint arXiv:2401.01787*, 2024.

[6] Peini Yi, Wenchi Cheng, Jingqing Wang, and Wei Zhang. RIS-assisted seamless connectivity in wireless multi-hop relay networks. *IEEE Transactions on Mobile Computing*, 2025.

[7] Jiaqing Chen, Pu Miao, and Kang Song. Multi-RIS-assisted covert communications with enhanced physical layer security in cooperative noma networks. *IEEE Open Journal of the Communications Society*, 2026.

[8] Neji Mensi and Danda B Rawat. On the performance of partial ris selection vs. partial relay selection for vehicular communications. *IEEE Transactions on Vehicular Technology*, 71(9):9475–9489, 2022.

[9] SM Saumik Shahriyer, ASM Badrudduza, Sarjana Shabab, Milton Kumar Kundu, and Heejung Yu. Opportunistic relay in multicast channels with generalized shadowed fading effects: A physical layer security perspective. *IEEE Access*, 9:155726–155739, 2021.

[10] I. S. Gradshteyn and I. M. Ryzhik. *Table of Integrals, Series, and Products*. Academic, San Diego, CA, 7th edition, 2007.